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 TOPICAL REVIEW

Multi-Crop Disease Detection in Computer Vision for Resource-Constrained Farms—A Review

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ABSTRACT Crop diseases remain a major threat to global agricultural productivity, particularly in resource-constrained regions, where early intervention is critical to ensuring food security. Smallholder farmers often practice intercropping, creating complex field conditions that challenge conventional disease detection systems. Although computer vision and machine learning offer powerful tools for early diagnosis, most existing approaches are tailored to single-crop scenarios and rely on cloud-based or high-resource environments, limiting their applicability in the field. This review synthesizes and explores recent advances in AI-based crop disease detection with emphasis on multi-crop and intercropped systems. We provide an overview of the detection pipeline from data acquisition and preprocessing to model training and deployment. We highlight technical challenges such as dataset scarcity, lack of modality diversity, poor model generalization, and limited real-world validation. Unlike prior reviews that isolate crops and modalities, this review uniquely integrates multi-crop, multimodal and edge deployable perspectives. We identify key research gaps and outline emerging opportunities for scalable solutions, including lightweight, multimodal models, offline-capable inference, edge deployment, and the integration of conversational AI for field-level support. By centering on the realities of low-connectivity, heterogeneous farming environments, this review aims to offer guidance toward inclusive and sustainable AI-driven agriculture.

INDEX TERMS Computer vision, crop diseases, crop disease detection, datasets, edge deployment, intercropping, lightweight models, machine learning, model deployment.

I. INTRODUCTION

In recent years, sustainable agriculture has gained attention as a means of ensuring long-term agricultural productivity and food security [1], [2]. However, agricultural sustainability heavily depends on effective disease management. Crop diseases are a major threat to sustainable agriculture because they reduce the efficacy of agricultural inputs and require intensive management [3]. Crop diseases destroy crop harvests if they are not monitored, detected, managed and controlled promptly [4], [5], [6]. The challenge of crop diseases is worsened by changes in climate, which increases

the prevalence and severity, further threatening agricultural sustainability [7].

Crop diseases have dire consequences for both the economic stability of agricultural communities and the availability of food at national and global levels, causing annual losses of approximately USD \$220 billion [8]. This global trend has been observed in Africa, Latin America and other regions around the world. In Tanzania, cassava blight streak disease-related losses were estimated to be over 860,000 tons, equivalent to more than USD \$51 million annually [9]. In Latin America and globally, crop diseases due to the presence of diverse plant pathogens and microorganisms cause economic losses every year [10].

Several important food crops in various countries include; cassava, beans, maize, potatoes, soybeans, ground nuts,

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millet, wheat, rice, corn, bananas, cashew nuts, yams, sorghum, etc. These crops are considered food security crops globally because of their resilience, ability to withstand droughts and soils with low-nutrient content [11], [12], [13], [14]. Over 800 million people worldwide, especially in developing countries depend on these crops as a source of calories, nutrients and income for livelihood [14], [15]. Most of these crops are usually intercropped with one another on the same field.

Intercropping is a common practice in tropical regions where various crops are grown simultaneously on the same field offering advantages to small scale farmers [16], [17]. In most cases, a field growing cassava has beans, maize and soybeans intercropped with it so as to improve soil fertility, reduce weed and soil erosion, provide diverse sources of income for farmers and enhance land productivity and food security [18], [19]. Studies show that intercropped fields generally demonstrate biological advantages over sole cropping [20]. However, when one crop gets infected with a disease, there are higher possibilities that another crop on the same field may get infected as well with any other disease. Different crops possess and exhibit unique physiological traits and disease symptoms, making it challenging to generalize models trained on specific crops to other crops.

Early crop disease detection is important for efficient and sustainable food production. This allows timely and prompt intervention, which can significantly reduce the potential damage that could be caused [21]. This also limits the wide spread of disease epidemics that can have catastrophic effects on food security globally [22]. With early crop disease detection, losses in yields are minimized which contributes significantly to the advancement of sustainable agriculture [23].

Recent advancements in technology have enabled the development of intelligent systems that use Artificial Intelligence (AI) based systems, Internet of Things (IoT), and remote sensing for accurate and timely crop disease detection [24]. These systems utilize sensors and data analysis techniques to monitor crop health [25]. In particular, machine learning and computer vision have been to some extent explored to assist with early crop disease detection. Different models have been developed to perform crop disease detection on different crops, although in most cases they are tied to a single crop.

Despite the fact that several researchers have made substantial progress towards early crop disease detection, this review aims to identify the opportunities and challenges in the field of multi-crop disease detection for resource-constrained farms. Most reviews focus on a single-crop setup contexts and thus focus on models or solutions that cater for only that crop and its disease yet the same field may have several other crops that may also need diagnosis for better disease management of the farm. Additionally, many reviews are tied to RGB imagery which has limitations in asymptomatic cases. In addition to that, they give little or no attention to deployment of the models in the real-world setup which leaves a gap between model training and model

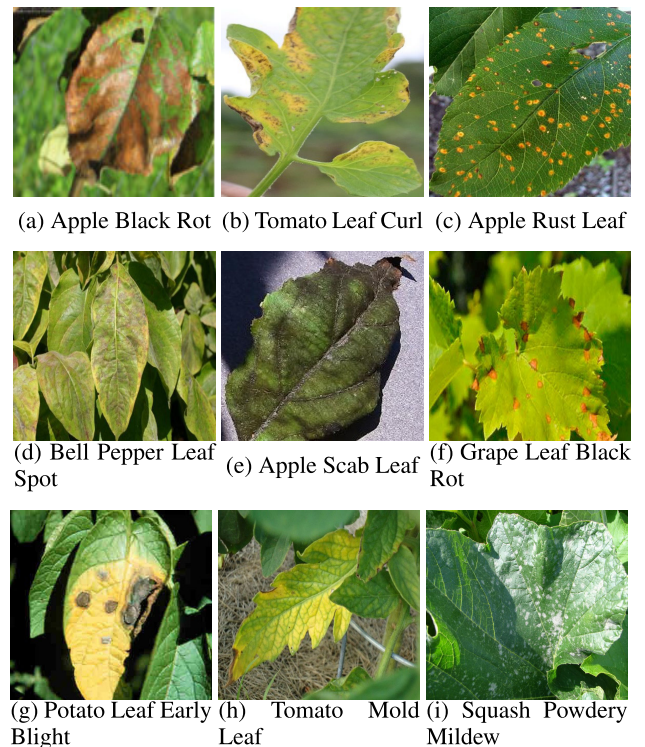


FIGURE 1. Sample diseased crop leaves from PlantVillage [26] and PlantDoc datasets [27].

deployment. Several reviews do not consider settings of resource-constrained farms for smallholder farmers where computation resources are limited requiring consideration of lightweight models. This review is distinct in its focus on multi-crop models for intercropped field scenarios, model deployment, different sensing modalities, multimodal datasets and approaches. It uniquely synthesizes recent developments across multimodal data modalities (e.g., spectral, thermal, fluorescence, LiDAR), edge and offline deployment strategies. By bridging the gap between model development and real-world deployment in low-connectivity agricultural contexts, this review provides a research direction for inclusive AI-driven crop disease management.

The review addresses the guiding question of: How can AI-driven crop disease detection be adapted to multi-crop, intercropped, and resource-constrained farm environments? The review is organized as follows; Section II describes traditional approaches and emerging technologies that highlight how farmers and researchers have historically detected and managed crop diseases, and how newer methods such as IoT, remote sensing, and computer vision are transforming detection practices. In Section III, we present the available data sets and data modalities. Later, in Section IV, we describe the model training process, with an emphasis on developing the ability to generalize between different crops and environments. In Section V, we discuss the deployment of the current designed model, with an emphasis on the trade-off between performance and field usability. Finally, we discuss opportunities and future directions in Section VI.

II. OVERVIEW OF CROP DISEASE DETECTION WITH COMPUTER VISION AND MACHINE LEARNING

A. CROP DISEASES

Crops are affected by a wide range of diseases, each posing significant challenges to their health and productivity. Crops are susceptible to fungal, bacterial, viral, nematode and parasitic diseases [28]. These crop diseases can damage the entire yields prior to harvesting if not monitored, detected, managed and controlled promptly [4]. Table 1 shows a number of common crops and the respective diseases that affect them and Table 4 lists the available crop datasets and their crop of interest. Fig. 1 shows a sample of diseased crop leaves for different crops extracted from the PlantVillage [26] and PlantDoc [27] datasets.

TABLE 1. Diseases affecting different crops globally.

Crop	Diseases
Maize	Maize Streak Virus (MSV), Southern Corn Leaf Blight, Gray Leaf Spot, Common Rust, Maize Dwarf Mosaic Virus (MDMV)
Cassava	Cassava Mosaic Disease (CMD), Cassava Bacterial Blight (CBB), Cassava Brown Streak Disease (CBSD), Cassava Green Mite (CGM), Cassava Leaf Blight
Beans	Anthraxnose, Rust, Common Bacterial Blight, Bean Common Mosaic Virus (BCMV), Bean Yellow Mosaic Virus (BYMV)
Cowpeas	Cowpea Mosaic Virus, Cowpea Rust, Bacterial Blight
Groundnuts	Early Leaf Spot, Late Leaf Spot, Rust, Collar Rot, Stem Rot, Powdery mildew
Soybeans	Soybean Rust, Soybean Mosaic Virus, Fusarium Root Rot.

B. TRADITIONAL APPROACHES TO CROP DISEASE DETECTION

Traditional crop disease detection relies on visual inspection, field surveys, and indigenous knowledge to identify visible symptoms such as leaf discoloration or shape deformation [29], [30]. Farmers and agricultural centers conduct surveys and sampling, often supported by extension workers who provide expert diagnosis and management advice. They also depend on traditional forecasting methods, using past seasonal patterns to anticipate outbreaks [31], [32], and monitor pest populations as early indicators of diseases [33], [34].

Environmental conditions such as rainfall, humidity, and high moisture levels are monitored by farmers, as they are closely associated with fungal infections [35], [36]. However, these approaches are time-consuming, error-prone, costly, and limited by the availability and expertise of professionals, making them unfavorable for large and resource-constrained farms [37]. Consequently, delays in detection often lead to significant losses. Modern AI-based and digital methods have been adopted to address many of these challenges [6], [38].

C. EMERGING TECHNOLOGIES IN CROP DISEASE DETECTION

Different studies present various technologies being applied in crop disease detection. The technologies include AI-based systems, remote sensing, IoT, genomics and molecular DNA-based pathogen detection techniques, among others.

These technologies are supported and enabled by a number of devices such as drones, digital cameras, mobile phones, and spectrometers. Fig. 2 shows some of the devices enabling emerging technologies in crop disease detection. Next, we present the emerging technologies highlighting the strengths and limitations of each and how they have been applied to different crop settings.

1) ARTIFICIAL INTELLIGENCE

Artificial intelligence (AI) techniques, particularly machine learning (ML) and computer vision, enable automated, accurate, and scalable crop disease detection by analyzing leaf images to identify subtle disease symptoms [39], [40]. Among ML architectures, Convolutional Neural Networks (CNNs) have shown high effectiveness in capturing complex visual and spectral features such as leaf spot size, color variation, temperature changes, and spectral reflectance, facilitating precise disease classification [41], [42], [43].



(a) Drone



(b) Spectrometer



(c) Mobile phone with a spectral add-on sensor



(d) Digital camera

FIGURE 2. Data acquisition devices: (a) Drones equipped with UAV technology, (b) Spectrometers have been used for detecting viral diseases [48], (c) Mobile phones with spectral add-on sensors [49], and (d) Digital cameras.

Recent advances integrate multimodal data sources, including soil composition, weather parameters, sensor data, and plant health metrics, enhancing the robustness of CNN-based models [44]. The integration of different modalities further improves model reliability in dynamic agricultural environments [45].

Moreover, ML frameworks can detect asymptomatic infections, allowing early intervention [46]. These models are continuously adaptable, retraining on updated datasets to generalize across new crop varieties and disease types, ensuring sustained performance and scalability compared to static, expert-dependent traditional methods [47]. In computer vision for crop disease detection, it is common to follow the steps described in Fig. 3, that are data acquisition, data

preprocessing, and model training and model deployment, discussed in Sections III, IV, and V, respectively.

2) REMOTE SENSING TECHNOLOGIES

Remote sensing technologies of satellites, drones and Uncrewed Aerial Vehicles (UAV) have been used for large scale monitoring by capturing spectral data, digital image data, fluorescence images and thermal images that can indicate crop health status such as changes in leaf pigmentation and moisture content [50], [51], [52], [53]. Remote sensing technologies can cover extensive areas and a wide coverage of fields. They are equipped with advanced sensors which offer finer visible, spatial and temporal resolutions, enabling early detection of diseases at the field level [54], [55], [56]. However, the devices like satellites may face challenges such as cloud cover, which can obstruct data capture, and the resultant resolution might be insufficient for detecting diseases at an early stage. Additionally, UAVs and drones tend to have restricted battery life, which can limit their ability to cover large areas in a single flight. Setting up the remote sensing systems involves high initial costs and requires training for effective operation and data interpretation. Several works have been done on crop disease detection using remote sensing technologies [51], [55]. From the analyzed studies, remote sensing has not been extensively explored for multi-crop disease recognition in intercropped fields, despite its potential advantages.

3) INTERNET OF THINGS AND WIRELESS SENSOR NETWORKS

Other works have explored IoT and wireless sensor network technologies in detecting crop diseases. They also focus on predicting conditions that favor crop disease outbreaks [57], [58], [59]. Data from several sensors in the environment, such as soil moisture, temperature and humidity, is integrated and transmits real-time updates to assist in early crop disease detection [60], [61]. IoT-enabled sensor networks enable continuous and real-time monitoring of environmental features. IoT systems enable farmers to access data remotely through cloud platforms. Alerts can be sent when environmental conditions are likely to lead to disease outbreaks, allowing for timely interventions [62]. Sensor networks can be scaled to cover large agricultural fields, providing comprehensive data across different areas for proper disease management. On the other hand, IoT devices collect a huge amount of data that requires robust data management and analytics. Poor data analysis can reduce the system's effectiveness in disease detection. The set-up of IoT systems including sensors and communication infrastructure, can be costly and this may limit smallholder farmers from adopting the technology [63], [64].

4) GENOMICS AND MOLECULAR DNA

Genomics and molecular Deoxyribonucleic Acid (DNA)-based Pathogen Detection techniques such as Polymerase

Chain Reaction (PCR) and Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) based diagnostics help in identifying pathogens at an early stage [65], [66]. This is made possible by breeding crop varieties resistant to diseases and identifying genetic markers associated with disease resistance. PCR and CRISPR-based diagnostics can detect very small amounts of pathogen DNA, ensuring early detection of diseases before symptoms appear. This reduces the risk of disease spread and enables timely interventions. Genomics and molecular DNA targets unique genetic sequences of pathogens, minimizing false diagnostics [67] and offering high specificity. They are versatile, provide fast results and can amplify DNA within a short time. CRISPR-based diagnostics offer results in even shorter time frames with simple setups. This can be a field for further exploration on different crops and conditions.

However, these techniques require sophisticated equipment and skilled personnel, which can be costly in resource-limited settings. These methods require high quality DNA samples otherwise, contaminated or degraded samples can lead to false negatives and unreliable results. Whereas advances are being made to create portable PCR and CRISPR kits, traditional implementations still rely on laboratory infrastructure. This remains an emerging field with limited application and exploration across multiple crops limiting their use in field conditions.

5) MOBILE APPLICATIONS AND DECISION SUPPORT SYSTEMS

Mobile apps can integrate with sensors and imaging technologies, allowing farmers to upload images of crops or input environmental data for instant disease diagnosis and suggestions on management practices. Many mobile applications are designed to be user-friendly, enabling even small scale farmers to access advanced diagnostics and recommendations via smartphones [68]. Compared to traditional lab-based methods, mobile apps provide a lower cost alternative for disease monitoring and management. They also offer community and knowledge sharing e.g Plantix, where users can share experiences, enhancing collective learning and support [69].

However, the mobile apps available and decision support systems require internet connectivity to operate and this brings in a need for infrastructure to facilitate the connectivity. Existing mobile applications like Plantix, Agro diagnosis tool, Nuru, Agrio etc. have restricted coverage and accuracy and might be restricted to specific regions or single crops. They can struggle with accuracy in diverse environmental conditions or less common disease variants. There would be great relevance if these mobile applications can incorporate several crops at a go so as to increase the effectiveness of their diagnosis on intercropped fields. Furthermore, these platforms for large scale model deployment and offline operation could address several existing gaps in this field of crop disease detection using machine learning.

TABLE 2. Emerging technologies in crop disease detection categorized into data acquisition, data analysis, and end-user delivery platforms.

General Category	Technologies	Strengths	Limitations
Data acquisition & monitoring	- Imaging & remote sensing (satellites, UAVs, spectral, thermal, fluorescence) - IoT & sensor networks (soil moisture, temperature, humidity sensors, WSN) - Genomics & molecular diagnostics (PCR, CRISPR, DNA sequencing)	- Captures diverse data sources (visual, environmental, genetic) - Enables early and large-scale monitoring - High specificity (genomics) and wide coverage (remote sensing)	- High cost and infrastructure requirements - Limited usability in resource-constrained settings - Requires skilled personnel and complex data processing
Data analysis & interpretation	AI & data-driven approaches (machine learning, deep learning, computer vision, multimodal fusion)	- Automates disease diagnosis - Scalable across large datasets - Adaptable to multiple crops and conditions	- Dependent on large and diverse datasets - Poor generalization across regions and intercropped fields - Risk of bias from dataset imbalance
End-user platforms & decision support	Mobile applications & advisory systems (cloud-based platforms, apps like Plantix, farmer dashboards)	- Affordable and user-friendly - Accessible via smartphones - Facilitates community knowledge sharing	- Often require internet connectivity - Limited accuracy in diverse conditions - Restricted to specific crops or regions

In Table 2, we present the advantages and disadvantages of emerging technologies categorized from the perspective of data acquisition, data analysis and end-user delivery systems. We can observe that no single technology is sufficient alone. The design of a system requires the integration of data acquisition (remote sensing, IoT, genomics), its analysis (AI/ML), and later deployment (mobile applications). Advanced tools like (spectral, genomics, IoT) provide higher accuracy but are expensive and infrastructure-heavy, while simpler tools (mobile apps, lightweight AI) are more accessible but provide lower performance.

D. EXISTING REVIEWS ON CROP DISEASE DETECTION

A review on machine learning for detection and prediction of crop pests and diseases is presented in [70] focusing on tasks of classification, detection, prediction of diseases and pests, with an emphasis on tomato crops. Focus is on image classification looking at variables that influence crop pests and diseases plus agricultural datasets. This survey does not look into the deployment of this model on edge devices leaving a gap between the model and the end users. Omaye et al. presents a machine learning approach focusing on plant disease detection for crops like apples, cassava, cotton, and potatoes [71]. Exploration is done on how ML algorithms are utilized not only for disease detection but also for integrated plant disease management strategies and trends. However, consideration for a multi-crop perspective in a single model is not considered yet it could reduce the issues of poor generalization.

A comprehensive overview of techniques for crop disease detection which categorizes detection methods into visual inspection, spectroscopy, image analysis, and sensor-based approaches is presented in [72]. The study emphasizes the

economic and environmental impacts of effective crop disease management. Adhinata et al. presents a review on weed and crop classification using machine learning emphasizing trends of precision agriculture and smart devices in weed management [73]. Consideration into the issue of crop disease recognition in particular has not been recorded in this study. A review on machine learning techniques used in crop yield prediction is presented assessing advanced techniques like deep learning [74]. Another study examined crop disease detection with deep learning, offering a performance analysis of machine learning techniques applied across different crops [75]. Li et. al in [76] discusses trends and unresolved problems while analyzing deep learning architectures and their application in plant disease detection.

A study in [77] provides state-of-the-art machine learning methods that use different data sources, applied to plant disease detection listing the traditional and deep learning methods associated with the main data acquisition modalities. Work on several deep learning methods used for different crop disease segmentation and classification is presented in [78] summarizing the available research on different diseases on various crops based on machine learning and deep learning. In [79], multiple deep learning algorithms are used to demonstrate the current solutions for different crop disease diagnosis.

Existing reviews on crop disease detection using computer vision present machine learning for disease detection applied to crops such as corn, cassava, rice, maize, soybeans, cucumber, pepper, sorghum and tomatoes [80], [81], [82], [83] among others. However all these present work on crop disease detection looking at a single specific crop in isolation at a time and not a focus on multiple crops at ago. To the best of our knowledge, crop disease detection

reviews on multi-crop models in the context of intercropping remain limited. Algorithms trained and optimized on data for one crop such as cassava may fail to accurately detect diseases on another crop such as maize or beans or soybeans leading to misclassification and reduced model performance. In general, visual and metabolite features that are relevant for detecting disease in one crop may not transfer well to another.

Several other reviews in [84], [85], [86], and [87] present studies on plant and crop disease detection, classification and imaging techniques. However most of these focus on RGB (visible light) image data and not much has been done on integration of different data modalities like spectral, thermal, fluorescent among others to maximize the advantages of each in isolation and also when combined together.

Furthermore, most of these reviews do not provide a complete cycle of the phases through to model deployment, evaluation, and beyond. These reviews have not explored the deployment phase of the model in integrated devices that farmers use in real fields. They lack review of the usability and deployment of these models on edge devices which is important for applicability. Models are developed and tested in virtual environments working well however, another important part of the process is deployment of these models on real devices for accessibility and usability to the farmers. Additionally on deployment, consideration of the resource-constrained farms and devices for model deployment need to be reviewed and looked into further.

Earlier reviews have emphasized disease detection and individual crops, but few have considered multi-crop that works well in complex intercropped fields. From the categorization of existing review works, it is revealed that majority of prior studies on crop disease detection are limited to single-crop contexts and rely almost exclusively on RGB (visible light) imagery as seen in Table 3. Although a few reviews like [70], [74], and [78], extend their scope to multiple crops, they primarily examine separate single-crop studies rather than simultaneous multi-crop detection frameworks. None of the identified reviews address the critical issue of model deployment or usability in real-world, resource-constrained environments, and only a few briefly mention non-visual or multimodal modalities such as spectroscopy or chemical sensing. This pattern presents a persistent gap in the literature of lack of integrative, end-to-end analyses that encompass multimodal sensing, multi-crop or intercropping scenarios, and lightweight deployment strategies. Hence, there remains a clear research opportunity to develop holistic frameworks that move beyond model benchmarking to practical, field-deployable, and inclusive AI-driven crop disease detection systems.

E. OUR REVIEW METHODOLOGY

This paper adopts a narrative review approach designed to provide an integrative overview of advances, challenges, and emerging opportunities in multi-crop disease detection for

TABLE 3. Existing review papers on crop disease detection in computer vision. To the best of our knowledge, none of these reviews investigates model deployment.

Review Work	Crop Focus	Multi-crop Disease Detection?	Describe Data Modalities?
Ngugi <i>et al.</i> (2024) [75]	Single-crop	No	RGB only
Omaye <i>et al.</i> (2024) [71]	Multi-crop	No	No
Abdul Kareem <i>et al.</i> (2023) [72]	Single-crop	No	RGB only
Patil <i>et al.</i> (2022) [78]	Multi-crop	Yes	No
Domingues (2022) [70]	Multi-crop	Yes	No
Bali <i>et al.</i> (2022) [74]	Multi-crop	Yes	No
Sharma <i>et al.</i> (2022) [79]	Single-crop	No	RGB only
Vasavi <i>et al.</i> (2022) [86]	Single-crop	No	RGB only
Khakimov <i>et al.</i> (2022) [36]	Single-crop	No	Chemical sensing
Abade <i>et al.</i> (2021) [82]	Single-crop	No	RGB only
Li (2021)	Single-crop	No	RGB only
Singh <i>et al.</i> (2020) [88]	General	No	RGB only
Martinelli <i>et al.</i> (2015) [87]	Single-crop	No	Spectroscopy only

resource-constrained farms. The review offers an integrative synthesis of research on crop disease detection, with a focus on computer vision.

Relevant works were identified through exploring scholarly sources that included; IEEE Xplore, Scopus, Web of Science, Science Direct, Springer Link, and Google Scholar. The keywords and search strings used were; “Crop disease detection”, “Computer vision for crop disease detection”, “Machine learning in crop disease detection”, “Edge device deployment in agriculture” and “Datasets for crop disease detection”. Searches were limited to articles published between 2014 and 2025, written in English, and focused on crop disease detection using machine learning and computer vision technologies. The results were narratively integrated, and the studies were compared and contrasted across different topics.

1) INCLUSION CRITERIA

Peer-reviewed journal articles, conference papers, internet and book chapters were used in this review. Studies focusing on crop disease detection using Machine Learning, Deep Learning, Computer Vision and IoT were considered. Research addressing datasets, models, resource utilization, and deployment techniques on resource-constrained devices was reviewed. Papers discussing edge device deployment of models were considered as well.

2) EXCLUSION CRITERIA

Non-English publications, articles lacking methodological clarity, and papers with incomplete data. Studies focusing on machine learning applications without focus on crop disease recognition insights were excluded.

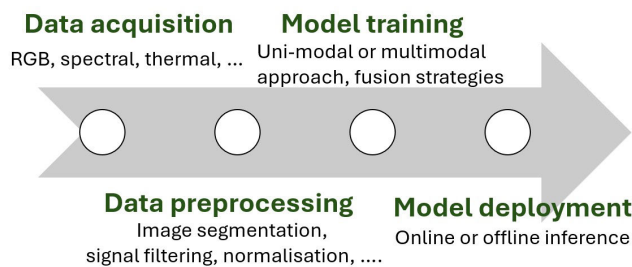


FIGURE 3. Standard protocol in a data analysis process.

III. DATA ACQUISITION, DATA PREPROCESSING AND DATASETS

The effectiveness of any crop disease detection system is fundamentally dependent on the quality and diversity of the data it uses. High quality and representative datasets are essential not only for training robust machine learning models, but also for ensuring that these models generalize well across multiple crops and diverse agro-ecological conditions. This section synthesizes how crop disease data are collected and prepared for training crop disease models, emphasizing their implications for multi-crop generalization and edge deployment. Table 4 presents a taxonomy of different sensing modalities, available datasets, the country of origin where the dataset was collected, the number of samples, and the crop diseases considered.

A. DATA ACQUISITION

Several image acquisition techniques have been explored to capture features and plant symptoms for early disease detection [88], [89]. Data can be collected manually or obtained from existing public datasets on the crop of interest. This can be spectral [90], visible light (RGB) images [83], thermal data, fluorescent data among others [26], [91]. Fig. 4 shows different data modalities in crop disease detection. Several methods used in data acquisition include Field Surveys, where manual collection of images directly from crop fields is done using smart phones, and cameras of different spectral ranges or other hand held devices [92]. Depending on the type of device used in the collection, different features of the crop leaves are captured including: leaf shape, color, texture, structure, spectral reflectance, spot patterns, temperature variations, Normalized Difference Vegetation Index (NDVI), chlorophyll indices, spectral signatures, morphological features, chlorophyll fluorescence, temporal and spatial features among others [93], [94]. On larger fields, devices are usually mounted on UAVs and drones to capture a wider range of the field whereas on relatively small fields, handheld devices are used. Other

data can be acquired through synthetic generation and data augmentation of existing datasets [95].

B. DATASETS

1) RGB-VISUAL IMAGE DATASETS

Digital cameras and smart phones are the most widely used devices in collection of the RGB visible light images [96], [97], [98]. They can be handheld for smaller fields or mounted on drones for larger field. The PlantVillage dataset [26] and PlantDoc dataset [27] are some of the widely used RGB image datasets in training machine learning and deep learning models for crop disease detection. There are several other custom datasets that are being used as seen in [43], [99], [100], and [101]. However most of the custom datasets most datasets contain images of single crops, with only a few covering multiple crop types. In addition to that, RGB images capture crop leaves exhibiting visual symptoms of different diseases like shape, color and texture, making them suitable for classification models based on visible light features [102]. However they have limited sensitivity to subtle physiological changes that show up before the visible symptoms. This presents a limitation for the detection of diseases in asymptomatic crops.

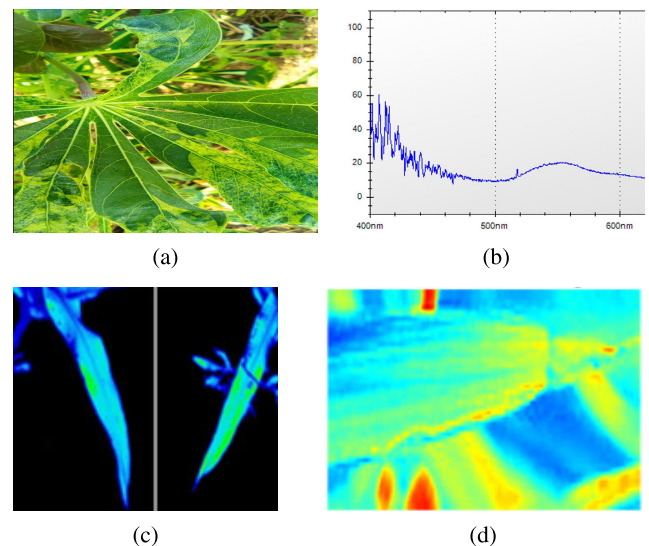


FIGURE 4. Different data modalities used in crop disease recognition: (a) RGB-visible light, (b) Spectral (c) Fluorescent, and (d) Thermal.

2) SPECTRAL IMAGE DATASETS

Spectral datasets usually contain images of either multispectral or hyperspectral reflectance. These images capture data across multiple wavelengths, including infrared and ultraviolet light, providing detailed information on crop health [48], [103]. They provide fine-grained data, distinguishing minute chemical or physiological differences and are useful for detecting early stress indicators, such as water stress or nutrient deficiencies, using wavelengths beyond visible light to detect subtle changes in plant physiology before visual symptoms appear [104], [105]. Multispectral images capture

TABLE 4. Datasets for crop diseases detection, crops considered, image type and the country of origin where the data samples were collected.

Modality	Dataset name	Image type	Crops	Number of images	Diseases considered	Country of origin	Link
RGB (Visible light)	Sugarcane Leaf(2024) [111]	RGB dataset	Sugarcane	6748	10 classes of viral diseases	India	No open access
	CCMT Multi-crop (2023) [99]	RGB (multi-crop)	Cashew, Cassava, Maize, Tomato	24881	18 classes of crop diseases	Ghana	CCMT Dataset
	Tomato Village(2023) [112]	RGB (field dataset)	Tomato	4526	Tomato leaf diseases for 8 classes	India	No open access
	Mak.Univ Beans Dataset(2022) [113]	RGB	Beans	15335	Bean Rust, Angular Leaf Spot	Uganda	Beans-dataverse
	Maize Image Dataset(2022) [114]	RGB	Maize	15821	Maize Streak Virus, Maize Leaf Blight	Uganda	Maize-dataverse
	Cassava Image Dataset(2022) [115]	RGB	Cassava	15000	CMD, CBSD	Uganda	Cassava-UG
	NMAIST Dataset(2022) [116], [117]	RGB	Bananas, Maize	34,345	Maize lethal Necrosis, Maize Streak Virus Black Sigatoka, Fusarium Wilt Race	Tanzania	NMAIST.dataverse
	KaraAgroAI Cocoa Dataset(2022) [118]	RGB	Cocoa	not specified	Cocoa Swollen Shoot Virus Disease and Anthracnose	Ghana	Restricted
	KaraAgroAI Maize Dataset(2022) [119]	RGB	Maize	15821	Maize Streak Virus, Maize Leaf Blight	Ghana	Agro-Maize
	NUST Maize Dataset(2022) [117]	RGB	Maize	9034	Fall Army Worm, Maize Streak Virus	Namibia	Maize-NUST
Thermal	PlantDoc (2020) [27]	RGB (field dataset)	13 crops	2569	17 crop disease classes	India	PlantDoc
	PlantVillage(2016) [26]	RGB (curated)	14 crops	54306	26 crop disease classes	USA	PlantVillage
	Paddy Doctor dataset(2022) [120]	Thermal & RGB images	Rice	16225	Not specified	India	Paddy-leaves
Spectral, Multi-spectral, Hyper-spectral	Thermal Persea(2024) [121]	Thermal imaging	avocado, acerola	15444	7 Classes of diseases	Spain	No open access
	Almond Thermal(2016) [122]	UAV thermal	Almond	not specified	Red leaf blotch	Spain	No open access
	Olive Thermal(2015) [123]	UAV thermal	Olive	not specified	Verticillium wilt	Spain	No open access
Spectral, Multi-spectral, Hyper-spectral	Ultra-High Hyperspectral(2024) [106]	Hyperspectral imagery	Cabbage, Eggplant	not specified	Stress + diseases	India	No open access
	Cassava Spectral(2023) [90]	Spectral dataset	Cassava	15,720 spectral samples	CMD, CBSD	Uganda	Cassava-spectral
	Olive Hyperspectral(2023) [46]	Hyperspectral dataset	Olive	not specified	Verticillium dahliae Pleospora herbarum	Greece	No open access
	Tea Spectral(2023) [94]	Spectral reflectances	Tea	not specified	Stress + foliar diseases	Indonesia	No open access
LiDAR	Hyperspectral(2023) [124]	Hyperspectral images	16 crops	385 spectral images	Viral infections	UK/Europe	No open access
	Crops3D(2024) [125]	LiDAR 3D scans	10 crops	Not specified	correlation with disease impact	China	Crops3D
	Pheno4D(2021) [126]	LiDAR temporal dataset	Maize, Tomato	224 point clouds	Not disease-specific	Germany-Bonn	Pheno4D
Fluorescent	TERRA-REF(2020) [127]	spectral, thermal, LiDAR	Sorghum, Wheat	not specified	Abiotic + biotic stress	USA	TERRA
	Tobacco Fluorescence (2025)	Fluorescence imaging	Tobacco	1000	Tobacco Mosaic Virus	China	No open access
	Citrus Fluorescence (2017) [128]	Chlorophyll fluorescence	Citrus	2184	Huanglongbing (HLB)	China	No open access
	Synthetic Fluorescence(2022) [129]	Synthetic + empirical fluorescence	Arabidopsis thaliana	108	Pseudomonas syringae pv. tomato	France	No open access
Fluorescent	CFIHL (2024) [130]	Chlorophyll fluorescence	hydroponic lettuce	not specified	Growth + stress phenotyping	China	CFIHL

spectral reflectance in visible and near-infrared wavelengths, while hyperspectral images capture spectral signatures.

Dedicated datasets composed of spectral images of crop leaves have been collected and used to train models for the early prediction of crop diseases before symptoms appear [48], [106], [107]. Machine learning algorithms such as K-Nearest Neighbors and Random Forest have shown promising results processing extracted descriptors from this data modality [105], [108]. Spectral data collection requires specialized equipment and technical expertise for analysis, making it costly. Although powerful, hyperspectral imaging can require significant computational effort and is often more expensive, making it more viable for research or controlled experiments than for large-scale monitoring [109], [110], creating a gap between technological potential and field reality.

3) THERMAL IMAGE DATASETS

Thermal images have been explored to a relatively lesser extent in the detection of crop diseases using computer vision. Thermal image datasets are developed using thermal cameras to capture temperature variations in plant canopies, which can be indicative of water stress or diseases [131], [132]. Thermal cameras measure leaf temperature to detect disease-related stress before visible symptoms appear, as sick plants often show temperature changes from disrupted transpiration. They can be handheld or drone-mounted for crop disease detection [133]. Thermal cameras capture spatial and temperature data from crops, allowing continuous monitoring of plant health over time [134], [135]. A limitation of this data modality is that thermal images are highly sensitive to environmental factors like temperature and humidity, which can introduce noise and affect model performance. Moreover, thermal imaging devices are costly, making datasets with such images rare. Further research is needed to optimize the use of thermal features across different crop types.

4) FLUORESCENCE IMAGE DATASETS

Fluorescence images measure the fluorescence emitted by chlorophyll under specific lighting conditions. They indicate photosynthetic efficiency, which can be affected by diseases [128], [136]. These fluorescent images require specific sensors and may be less practical for large-scale field deployment. The availability of such datasets is very limited, as collecting fluorescent images is costly and has been explored only to a small extent. To overcome this limitation, researchers have generated synthetic fluorescent images of diseased plants [130].

5) OTHER DATASETS

Several other datasets exist in the field of crop disease detection, although they are fewer in number. LiDAR (Light Detection and Ranging) technology, which uses laser scanning to create detailed 3D models of crop structures, has been explored in [125] and [126]. These datasets can support structural analysis, such as estimating canopy density,

which may correlate with disease severity. The disadvantage with LiDAR technology is the high cost and complexity of data interpretation. Climate weather datasets have been explored since certain environmental conditions facilitate the breeding of pests and diseases. They are used in textual and tabular form to complement image-based analysis by providing context for predictions. However, they require integration with sensor networks and reliable data collection which becomes complex and costly.

The survey of existing datasets indicates that whereas RGB image collections remain the most widely used due to their accessibility and cost effectiveness, they are limited to symptomatic visual cues. Advanced modalities like spectral, thermal, and fluorescence imaging offer earlier detection of physiological stress and diseases. However, their adoption is limited by high equipment costs, computational needs, and limited public availability of their datasets. Other data sources such as LiDAR and climate data provide valuable insights but require careful integration to be effective. Overall, the survey shows a high reliance on RGB data in the field. There is need for diverse, multimodal, and open-access datasets that represent multiple crops and varied field conditions. Such datasets are essential to train models that generalize well and remain effective when deployed in real-world smallholder farms.

C. DATA PREPROCESSING

Effective data preprocessing is crucial in preparing datasets for machine learning-based crop disease detection. Raw datasets often contain inconsistencies such as blurred images, poor lighting, or distortions caused by different environmental conditions. Preprocessing improves image quality and consistency, improving the accuracy and reliability of trained models. The most common preprocessing techniques include image resizing and normalization to maintain consistent dimensions and pixel ranges, contrast enhancement to improve the visibility of disease symptoms, and noise reduction to remove artifacts or irrelevant features that may hinder model training [97], [137]. Data augmentation techniques such as rotation, flipping, and scaling which help increase dataset diversity and improve model robustness against variations in lighting, orientation, and image quality [138], [139].

Methods such as filtering, geometric transformations, and mathematical morphology are used to refine image structure before features are extracted. Feature learning is further improved using algorithms such as Principal Component Analysis (PCA) and autoencoders for dimensionality reduction and anomaly detection [48], [140]. Additionally, Gaussian Mixture Models (GMM) have been used to model underlying data distributions for unsupervised preprocessing tasks. Models trained on inadequately preprocessed images tend to exhibit lower predictive performance, particularly when deployed on mobile or edge devices that are more sensitive to lighting, shadow, and angle variations. Proper

preprocessing not only improves model accuracy but also increases the reliability of real-time field applications.

1) IMAGE SEGMENTATION

Image segmentation divides images into meaningful regions, such as separating leaf areas from the background. Targeted segmentation enables more accurate feature extraction and contributes to higher model performance in classification tasks [141]. Several segmentation techniques have been explored for agricultural imagery, including deep learning methods such as U-Net and EfficientNet, encoder-decoder architectures, multi-scale approaches, dilated convolutional networks, visual attention models, generative adversarial networks (GANs), graph neural networks (GNNs), instance segmentation networks, and transformer-based models [142], [143]. U-Net hybrid models have demonstrated high segmentation accuracy, particularly in localizing leaf diseases [144]. Fully Convolutional Networks and Mask R-CNN are used to generate precise pixel-level masks for lesions [145]. Classical techniques of thresholding, region growing, K-means clustering, watershed algorithm, graph-based segmentation, and conditional random fields (CRF) offer simpler alternatives, especially when computational resources are limited [107].

In practice, segmentation pipelines often consist of multiple stages. For example, beginning with K-means clustering for initial segmentation, followed by gradient computation and region adjacency graph construction to refine boundaries. Segmentation quality is significantly affected by the choice of color space [146]. When effectively applied, segmentation techniques produce detailed, pixel-level classification masks that enable disease detection and classification, thereby improving overall performance of models [144].

D. MULTIMODAL DATA FUSION FRAMEWORKS

Multimodal methods combine diverse datasets such as RGB, spectral, thermal, fluorescence, and LiDAR imaging. They utilize their complementary strengths to achieve more accurate and robust crop disease detection. Fusion can be done at the data, feature, or decision level, improving generalization and allowing for early stress detection [147]. Recent studies reveal that such integration enhances diagnostic accuracy across varying conditions [148], [149].

To enable real-world deployment, research increasingly focuses on efficiency through TinyML, TensorFlow Lite, MobileNet, and EfficientNet Lite architectures. Quantization, pruning, and knowledge distillation reduce model complexity with minimal accuracy loss [150]. Edge deployment running trained models locally on devices such as smartphones or micro controllers supports low-power, offline operation and scalability in resource-constrained settings.

Combining multimodal fusion with edge-suitable strategies offers an opportunity to inclusive crop disease detection. Table 5 summarizes representative fusion strategies for building context-aware frameworks. These approaches

TABLE 5. Summary of fusion strategies for multimodal frameworks, their levels of integration & suitable deployment.

Fusion Type	Level	Description	Typical deployment
Single Modality	None	Uses a single input modality (e.g., RGB only) for training and inference	Edge
Early Fusion	Data level	Concatenates raw multimodal inputs (e.g., RGB + spectral + thermal) before feature extraction in a shared encoder.	Edge
Mid-Level Fusion	Feature level	Combines intermediate feature embeddings from different encoders, enabling modality-specific feature learning before fusion.	Edge+Cloud
Late Fusion	Decision level	Integrates independent model outputs (e.g., averaging, weighted voting, stacking) after separate inference pipelines.	Edge+Cloud
Hybrid Fusion	Adaptive/attention-based	Dynamically integrates both feature-level and decision-level cues through attention or ensemble mechanisms.	Edge + Cloud Hybrid

enhance robustness to environmental variability and enable earlier disease identification while remaining practical for smallholder use.

Regarding asymptomatic detection, spectral (especially hyperspectral) and fluorescence modalities are most effective before visible symptoms appear, whereas thermal and IoT data show partial correlations through stress indicators (Table 6). RGB remains the most accessible modality, but its fusion with hyperspectral or IoT data improves early detection. However, higher asymptomatic sensitivity often increases model complexity, challenging edge feasibility.

IV. MODEL TRAINING FOR CROP DISEASE DETECTION

The training stage typically involves splitting datasets into training, validation, and test sets, followed by model selection, hyperparameter tuning, and iterative model training and validation. Some of the machine learning and deep learning algorithms commonly used include Convolutional Neural Networks (CNNs), K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Decision Trees, Gradient Boosting methods (e.g., XGBoost), Naive Bayes, Long Short-Term Memory networks (LSTM), and Random Forests, among others [71], [151], [152]. CNNs have emerged as the dominant architecture for image-based crop disease detection because of their ability to learn spatial feature hierarchies from images [146], [153], [154]. Studies have shown CNNs to be effective in identifying visual disease symptoms such as leaf discoloration, spot distribution, lesion patterns, and shape anomalies [155]. CNNs can analyze complex crop features such as color variations, texture patterns, lesion shapes, and spatial arrangements of disease symptoms [41], [43], [156].

TABLE 6. Comparison of data modalities by their characteristics and edge deployment suitability.

Modality/ data type	Status of availability	Data acquisition cost	Features captured	Typical model size	Suitability for fusion	Suitability for compression	Suitability for edge deployment	Works well in asymptomatic conditions?
RGB (Visible Light)	Very high	Low (standard digital cameras/smartphones)	Color, texture, shape	Light–moderate 10MB–100MB	High	High	Excellent	No , detects only visible symptoms
Multispectral / Hyperspectral	Limited public access	Very high (specialized sensors/UAVs)	Reflectance across narrow spectral bands	Heavy (100MB – 1GB+)	Very high	Moderate (requires PCA/autoencoders)	Poor–moderate	Yes , captures biochemical stress before visible signs
Thermal imaging	Moderate; (UAV/greenhouse datasets)	Medium (infrared cameras)	Leaf temperature, transpiration rate	Light 20MB–80MB	Moderate	High (compact image sizes)	Good	Partially , detects stress but not pathogen specificity
Fluorescence imaging	Low; mostly research-level datasets	High (fluorescence sensors)	Photosynthetic activity, chlorophyll response	Moderate 100MB–200MB	High	Moderate	Fair	Yes , detects metabolic changes before visible symptoms
LiDAR (3D scanning)	Rare; limited open datasets	Very high (UAV/terrestrial scanners)	3D structure, canopy height, density	Very heavy 0.5GB–2GB	Moderate	Low (heavy point-cloud data)	Poor	No , structural changes appear late
IoT-sensor networks	Increasingly available (custom setups)	Variable (depends on scale)	Environmental conditions; soil, temperature	Very light (<10 MB)	Very high	Very high (tabular data easily compressed)	Excellent	Partially , detects stress conditions, not direct infection

Model performance in crop disease detection is often constrained by small or imbalanced datasets. To overcome these limitations, transfer learning has been widely adopted, enabling the fine-tuning of pre-trained models such as ResNet, MobileNet, DenseNet, and EfficientNet [6], [157]. This strategy accelerates convergence, enhances generalization, and reduces training time, particularly in data-scarce or resource-constrained settings [158]. In parallel, custom architectures have been developed to address domain-specific challenges such as multi-disease or multi-crop recognition, where diverse symptom patterns demand tailored network designs.

While deep learning architectures dominate image-based disease detection, classical machine learning algorithms including Support Vector Machines (SVM), Random Forests, K-Nearest Neighbors (KNN), and Decision Trees remain relevant for structured or tabular data, such as spectral indices, environmental sensor readings, or low-resolution imagery. Moreover, ensemble methods like Gradient Boosting (e.g., XGBoost) have shown competitive performance in predicting early disease severity from handcrafted features [146].

Model selection is typically task-dependent and guided by computational constraints and deployment goals. Lightweight models such as MobileNet are favored for on-field edge applications, where efficiency and low power consumption are critical, whereas high-capacity architectures like EfficientNet are suited for centralized, computation-rich environments. In complex agricultural settings, custom or hybrid networks are often proposed to handle multiple diseases or crop types simultaneously, achieving higher adaptability and robustness.

Ultimately, model success depends not only on architecture and training strategy but also on data quality and alignment with real-world deployment requirements. Training models for multi-crop disease detection often leads to heavy architectures that must be compressed to fit resource-constrained

devices. This motivates the next section, which focuses on model deployment. Table 6 summarizes model complexity across modalities, data availability, and acquisition costs and factors that can guide researchers in selecting appropriate configurations for practical field applications.

V. MODEL DEPLOYMENT

After a framework has been designed, tested, and validated, the next step is deployment to ensure its applicability and usability in real-world farming environments. Recent research has increasingly focused on edge deployment, aiming to reduce latency and computational demands while enabling real-time decision making [159]. This involves integrating disease detection models into smart agriculture platforms, where edge devices such as Raspberry Pi or mobile applications host the inference locally [160].

While cloud-based deployment remains common offering scalable and centralized processing its reliance on stable connectivity and associated costs often limits accessibility for smallholder farmers in rural areas. In such contexts, edge deployment provides a practical alternative, enabling offline operation and faster response times. However, edge devices are inherently resource-constrained, highlighting the need for lightweight, optimized models tailored to their limited computational and memory capacities.

Researchers emphasize the importance of tuning model recall and evaluating the influence of different input data types, such as images and video, to ensure compatibility with resource-constrained devices. Several deployment strategies, such as partitioning, quantization, and early-exit mechanisms, have been explored to optimize latency and performance across mobile, edge, and cloud platforms [161]. Techniques like kernel optimization, quantization, and the integration of Vision Processing Units (VPUs) have been successfully applied on devices such as the Raspberry Pi to reduce computational load and inference time [159].

TABLE 7. Lightweight models for edge deployment, their strengths and limitations.

Model Family	Description	Strengths	Limitations
Lightweight CNNs (e.g., MobileNet, EfficientNet-Lite, SqueezeNet)	Shallow or depthwise CNNs optimized for low-latency inference on mobile and edge devices.	Small size, fast inference, proven in agriculture apps.	May struggle with visually similar symptoms across species; limited capacity for complex, mixed scenes.
Quantized / Pruned Models (e.g., ResNet50 8-bit, Pruned EfficientNet)	Models compressed via quantization (e.g., int8) or pruning of redundant weights.	Reduced memory footprint, lower power and faster inference; suitable for very constrained devices.	Risk of accuracy drop, retraining or calibration may be needed.
Edge Transformers (e.g., MobileViT, Tiny-ViT, EdgeFormer)	Transformer architectures optimized for edge deployment using attention and convolutional hybrids.	Better spatial reasoning; powerful for mixed, occluded scenes.	Less explored in agricultural contexts; needs larger datasets and careful tuning for species diversity.

TABLE 8. Edge devices for lightweight model deployment, their memory size, cost and model types supported.

Edge Device	Memory (RAM)	Power Energy Use	Cost (USD)	Model Types Supported	Applicability
Mobile phones (smartphones)	4–12 GB	3–6 W (battery powered)	\$150–\$800	MobileNet, EfficientNet-Lite, Tiny-ViT, CNNs via TensorFlow Lite	Highly practical and accessible for farmers; supports on-device or hybrid inference.
Raspberry Pi 4 model B	4–8 GB	~5 W (5V × 1A)	\$60–\$90	MobileNet, EfficientNet-Lite, pruned CNNs via TensorFlow Lite	Suitable for small-scale inference; performs well with RGB datasets; limited for large multimodal or transformer-based tasks.
NVIDIA Jetson Nano	4 GB	5–10 W (active mode)	\$120–\$160	Quantized CNNs, Tiny-ViT, EfficientNet-Lite	High inference speed and accuracy; supports multimodal fusion; power-intensive.
Google Coral Edge TPU Dev board	1 GB shared	2–4 W	\$100–\$150	Quantized (int8) CNNs, MobileNet-EdgeTPU	Excellent power-performance balance; ideal for real-time, solar-powered field deployment; efficient for low-latency multi-crop classification.
Intel Neural Compute Stick 2 (NCS2)	Host-dependent	3–4 W	\$80–\$120	Quantized CNNs, SqueezeNet, MobileNet-SSD	Portable accelerator for Raspberry Pi; suitable for on-farm monitoring; moderate speed in complex multi-crop images.
ESP32/Arduino Nicla Vision	< 2 MB SRAM	< 1 W	\$10–\$30	TinyML, pruned CNNs, binary classifiers	Ultra low-cost, energy-efficient, suitable for simple disease alerts or sensor triggers; limited feature capacity for multi-crop models.
STM32H7 edge AI board	2 MB SRAM	0.8–1.5 W	\$25–\$50	TinyML, SqueezeNet-Micro, simplified CNNs	Excellent for ultra low-power, insufficient memory for diverse or high-resolution multi-crop datasets.

These efforts highlight the growing focus on efficient model deployment in computer vision applications across multiple domains. More recently, TinyML has emerged as a promising approach, enabling local data processing and model inference directly on resource-limited IoT devices, thereby reducing reliance on cloud infrastructure [150].

A. EFFICIENT AI MODELS FOR EDGE DEPLOYMENT

Recent research has focused on deploying lightweight machine learning models on edge devices to enable efficient, low-latency inference without relying on cloud connectivity. For example, MobileNet, a compact deep learning architecture, has been used for non-intrusive load monitoring on smartphones, with TensorFlow Lite further compressing the model to reduce memory usage and computation time [121], [162]. Studies have shown that models deployed at the edge can achieve faster response times than cloud-based systems while maintaining comparable accuracy, demonstrating

the potential of edge AI for real-time and secure applications [163].

To address the strict resource constraints of embedded systems, researchers have explored lightweight CNNs, quantized and pruned architectures, and emerging edge-optimized transformers such as MobileViT, Tiny-ViT, and EdgeFormer. These models capture long-range dependencies efficiently through compact self-attention mechanisms. Table 7 summarizes representative lightweight models, highlighting their key advantages and trade-offs for practical deployment.

B. EDGE DEVICES FOR MODEL DEPLOYMENT

Edge devices vary widely in computational capacity, cost, and energy demand. Recent studies have evaluated platforms such as NVIDIA Jetson and Raspberry Pi for machine learning applications, analyzing trade-offs in accuracy, power efficiency, and inference time [164]. Embedded machine learning has been applied to various agricultural tasks,

including crop disease recognition, classification, and diagnosis [165]. Mobile applications represent a common deployment channel for these models [166]; however, real-world performance often declines under uncontrolled conditions, such as variable lighting and orientation [151]. Despite these limitations, mobile-based solutions have proven valuable for large-scale adoption, particularly among farming communities embracing digital tools [167]. Table 8 summarizes commonly used edge devices, outlining their memory capacity, cost, and supported model types for lightweight deployment.

Deployment strategies for crop disease detection need to trade-off and balance between computational power, accessibility, and practicality. Cloud-based approaches provide high accuracy and scalability but are often impractical for smallholder farmers due to limited connectivity and high latency. Edge deployment on devices such as Raspberry Pi or NVIDIA Jetson, which brings inference closer to the field, reducing reliance on the internet, though constrained by limited resources. Mobile applications remain the most accessible solution, offering intuitive interfaces and broad reach, yet they may under perform in offline or low-light conditions. TinyML and micro controller-based systems offer ultra low-cost, low-power alternatives suitable for rural environments, enabling offline inference where data can be processed locally without network access. However, their restricted memory and compute capacity limit model complexity. Emerging hybrid architectures that integrate cloud, edge, and mobile layers are promising, combining real-time diagnosis with continuous model updates through cloud-based retraining. To meet these diverse deployment demands, researchers have proposed lightweight CNNs, quantized, and pruned models that reduce computational footprints while maintaining performance. These developments set the stage for the next section, which discusses the open challenges and research opportunities in deploying AI-based crop disease detection at scale.

VI. OPEN CHALLENGES AND RESEARCH OPPORTUNITIES

Several limitations in previous studies, along with emerging trends and future research directions, have been identified. This section summarizes these insights, emphasizing the development of lightweight models suitable for deployment on embedded devices that operate under constrained computational and energy conditions [6].

A. CURRENT LIMITATIONS

Although machine learning and computer vision have shown great promise for early crop disease detection, several limitations still hinder their practical deployment and effectiveness in real-world farming environments.

1) GENERALIZATION AND REGIONAL ADAPTABILITY

Models trained on datasets from one geographical region may fail to generalize well in another region due to differences in soil type, climate, crop varieties and farming practices.

An example is that a model trained on East African crop datasets may not perform as expected in West Africa due to differences in crop features and varieties [168], [169]. Models trained in controlled environments like lab settings may not perform as expected in real-world scenarios where these conditions are inconsistent because they lack robustness. This highlights the need for domain adaptation strategies, data augmentation, and inclusion of geographically diverse datasets in training pipelines [139].

2) CONNECTIVITY AND DEPLOYMENT CONSTRAINTS

The heavy reliance on cloud-based models and architectures limits the deployment of machine learning models in remote rural areas with poor or no internet connectivity. Cloud-based models require stable internet access for inference and relevant infrastructure which is often unreliable in remote farming regions. For example, a mobile app requiring cloud-based inference might fail in offline mode. Real-time inference in such areas is unreliable and limited by poor internet connectivity making it critical to shift toward edge and offline inference capable architectures. Techniques like use of TinyML frameworks can support development of lightweight models for deployment on low-cost, low-power devices like micro controllers and smartphones [160].

3) DATASET GAPS AND MODALITY IMBALANCE

Most publicly available datasets rely on RGB images, which primarily capture symptomatic disease stages. Visual symptoms of different crop diseases often overlap, leading to misclassification. However, early detection often requires other modalities like spectral, thermal, or hyperspectral imaging. Whereas these modalities offer improved detection capability especially in asymptomatic stages, their datasets are scarce and often not publicly available. There is a need for open-access, well-annotated multimodal datasets that cover diverse crops, diseases, and real-world field conditions. Multimodal fusion approaches can combine features from various data types to improve model robustness and effectiveness in early crop disease detection. This presents research opportunities to explore and develop datasets for several other data types other than RGB type and make them public. Further more, there is need to design multimodal models that integrate complementary features from different data sources, enabling richer representations and improved disease detection performance.

4) USABILITY AND HUMAN-CENTERED DESIGN

Many smallholder farmers lack technical knowledge to operate mobile or web-based crop disease detection tools. Therefore, they are often discouraged by complex mobile applications with unintuitive interfaces. Issues such as language barriers, the complexity of uploading images, and interpreting the outputs of the system further reduce the adoption of the technology. Therefore, future systems need to prioritize simplicity, visual guidance, and support

for local languages to improve accessibility and user engagement [170].

5) COST AND INFRASTRUCTURE BARRIERS

Setting up end-to-end computer vision systems including data acquisition hardware like drones, sensors, and high-performance processors is often cost prohibitive for smallholder farmers. Moreover, lack of power infrastructure and digital infrastructure compounds the challenge. Shared resource models, public subsidies, and low-cost alternatives like mobile-based solutions should be explored to reduce the digital divide in agriculture [169].

6) INTERCROPPING AND MULTI-CROP DISEASE DETECTION NEEDS

Most ML models are developed for monocrop settings, yet intercropping is prevalent in many developing regions. A single model that can detect diseases across multiple crops within the same field is still lacking. We foresee a clear opportunity to develop multi-crop classification frameworks that can operate effectively in such heterogeneous environments.

7) LACK OF END-TO-END PIPELINES

Many studies focus heavily on model development but fail to report on deployment, usability testing, or long term field performance. The absence of complete end-to-end pipelines from data acquisition to edge deployment limits the real-world impact of these systems. Rigorous testing in live farming conditions is necessary to understand operational challenges and iterate solutions effectively.

Data privacy concerns related to sharing crop and farm imagery where farmers may be hesitant to share sensitive agricultural data. Drone imagery could unintentionally capture neighboring farms. Implementing strict data privacy policies and anonymize datasets [170].

B. EMERGING TRENDS

Recent research is increasingly focusing on several key directions, including:

- Development of lightweight, edge-deployable models using MobileNet, EfficientNet, and TensorFlow Lite. Pulling inference from the cloud to the edge to support offline inference [162], [171].
- Exploration of multimodal fusion techniques combining RGB, spectral, and thermal imagery to improve accuracy, effectiveness, efficiency and robustness of models for early crop disease detection [148], [149].
- Building multi-crop, intercropping-aware classification systems [16].
- Integrating computer vision with Large Language Models (LLMs) for multilingual support and conversational interfaces [172], [173].
- Collaborative open access dataset development using drones, mobile phones, and IoT platforms [59].

- Use of Tiny Machine Learning (TinyML) architectures deployed at edge devices to bridge the gap of poor infrastructure and lack of connectivity [150]. TinyML offers a more accessible solution for crop disease detection, allowing ML models to run on low-cost, low energy devices [174].

These areas present opportunities to develop scalable, farmer-centric crop disease detection solutions. Consequently, emerging research can focus on embedded multimodal machine learning systems and multi-crop models capable of operating effectively in intercropped fields, aiming to create integrated solutions suited for diverse and resource-constrained farming environments.

C. OPPORTUNITIES AND FUTURE RESEARCH DIRECTION

Future research should aim to overcome the current limitations by focusing on the following directions.

1) MULTI-CROP AND INTERCROPPED FIELD MODELS

Most existing machine learning solutions are designed for single crop environments. There is need for development of models that can accurately detect diseases across multiple crops to support smallholder resource-constrained farming systems. Future research should focus on designing models capable of diagnosing diseases across several crop types and disease classes within the same inference pipeline. Diagnosing diseases across different crops in the same inference pipeline can cater for intercropped contexts that reflect real-world farming practices in smallholder farms.

2) MULTIMODAL DATA FUSION

Combining different data modalities such as RGB, spectral, thermal, and fluorescence imaging can improve the robustness and accuracy of disease detection. Multimodal fusion enables early identification of asymptomatic infections and provides richer contextual understanding of disease severity. Future research should focus on developing model architectures and training strategies optimized for multimodal inputs, moving beyond RGB imagery to incorporate complementary modalities. By effectively combining diverse data sources, these models can improve early detection capabilities in real-world agricultural settings.

3) LIGHTWEIGHT AND EDGE-COMPATIBLE MODELS

To enable practical deployment, models must be optimized for edge devices through techniques that reduce computational and memory needs while maintaining accuracy. Approaches such as model pruning, quantization and knowledge distillation can be adopted. Efficient architectures for example MobileNet, EfficientNet Lite, and TinyML, should be further explored to support edge deployment. Emphasis should be placed on balancing efficiency and performance to ensure reliable operation on low-power, embedded systems commonly used in real-world agricultural environments.

4) OFFLINE INFERENCE-CAPABLE APPLICATIONS

Given the connectivity limitations in rural agricultural regions, offline disease detection is crucial. There is need to develop mobile and embedded applications that perform on-device inference without internet access. Frameworks like TensorFlow Lite or TinyML can be used on devices for real-world usability.

5) DATASET DIVERSITY

Current datasets are limited in modality, geographic coverage, and crop disease representation. There is a pressing need for publicly available, annotated, and diverse datasets covering different crop varieties, different sensing modalities, growth stages, and environmental conditions. Collecting data with different devices and from different communities can support this goal to aid future developments.

6) USER-CENTERED TOOLS

Technical tools should accommodate users with limited digital skills. Interfaces should support multilingual input, visual guidance, and intuitive processes. Adopting inclusive design approaches to ensure accessibility and adoption where tools are co-designed with farmers. This can help to ensure ease of usability, multilingual support, and actionable decision outputs enhancing adoption and sustainability of AI-driven solutions in communities.

7) INTEGRATION OF LARGE LANGUAGE MODELS (LLMs)

Combining computer vision models with LLMs can support conversational, context-aware disease diagnosis and provide actionable recommendations. However, challenges such as model size, domain-specific tuning, and offline usability must be addressed to include LLMs effectively in farmer set-ups.

8) FIELD VALIDATION AND REAL-WORLD DEPLOYMENT

There is a notable gap between laboratory results and real-world performance. Future research should emphasize end-to-end deployment, pilot testing with farmers, and performance evaluation in varied field conditions to assess usability, reliability, and scalability.

By addressing these research directions, future innovations in crop disease detection can become more inclusive, scalable, and impactful, especially for smallholder farmers in resource-constrained regions.

VII. CONCLUSION

Crop disease recognition using computer vision and machine learning presents potential solutions to the persistent challenges faced by smallholder farmers, especially in regions prone to food insecurity. Whereas numerous techniques and frameworks have been proposed, their practical application in real-world farming contexts particularly in intercropped and resource-constrained environments remains limited. This review has identified key challenges including dataset scarcity and lack of diversity, limited support for multi-crop

detection, the absence of real-time offline capabilities, and high barriers to adoption due to infrastructure and usability issues. Moreover, most existing systems are designed for cloud-based environments and single crop scenarios, making them unsuitable for the heterogeneous, low-connectivity settings typical of many rural agricultural areas.

Emerging opportunities such as lightweight models for edge deployment, multimodal data fusion, and the integration of AI with IoT and mobile tools offer a direction to bridge these gaps. In addition, conversational AI systems that use large language models have the potential to offer contextual and multilingual support that is directly accessible to farmers. To fully realize this potential, future research must prioritize real-world validation, inclusive design, and suitable deployment strategies. Addressing these priorities will enable the development of robust, scalable, inclusive and accessible crop health monitoring systems that directly support sustainable agriculture and food security.

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